CENG 252: Chemical Reaction Engineering

Practice Midterm 2 (Lectures 1–10)

Time: 1 hour 20 minutes • Total: 100 points

Instructions

- Closed book, closed notes, no cheat sheet. A formula sheet is provided on the last page. Only a simple scientific calculator.
- **Show all work.** Write the mole balance, rate law, stoichiometry and energy balance before plugging in numbers. For mechanisms, write each step's rate law, put the fast steps at equilibrium (or apply the PSSH), and write the site (or area) balance before simplifying. Partial credit is given for a correct setup.
- State every assumption. Box your final answers and include units.
- The suggested times add up to about 65 minutes, which leaves about 15 minutes to check your work.

Problem	Topic	Points	Suggested time	Score
1	Energy balance (adiabatic CSTR)	25	18 min	
2	Reaction pathways and enzymes	20	13 min	
3	Cumene decomposition (§10.3)	30	20 min	
4	CVD of germanium (§10.5.2)	25	14 min	
	Total	**100**	65 min	

Constants: $R = 8.314$ J/(mol·K) $= 1.987$ cal/(mol·K) $= 0.08206$ dm^3·atm/(mol·K).

Problem 1: Energy balance: adiabatic CSTR

[25 pts]

The elementary, reversible liquid-phase isomerisation

$$\mathrm{A} \rightleftharpoons \mathrm{B}, \qquad -r_A = k\left(C_A - \frac{C_B}{K_C}\right)$$

is carried out adiabatically in a CSTR. The feed, 125 mol/min, contains 80 mol% A and 20 mol% of an inert solvent I. It enters at $T_0 = 300$ K with $C_{A0} = 2.0$ mol/dm^3 (no B in the feed).

Heats of formation at 298 K	$H_A^\circ = -35{,}000$ cal/mol, $\quad H_B^\circ = -41{,}000$ cal/mol
Heat capacities	$C_{P,A} = C_{P,B} = C_{P,I} = 40$ cal/(mol·K)
Kinetics	$k = 0.005$ min^{-1} at 300 K, $\quad E = 15{,}000$ cal/mol
Equilibrium	$K_C = 30$ at 300 K

(a) [3] Calculate ΔH_{Rx}° and ΔC_P. Does the heat of reaction change with temperature here? Why?

(b) [6] Starting from the steady-state energy balance on the formula sheet, derive the adiabatic relation between T and X for this feed (including the inert), and evaluate it numerically.

(c) [10] Find the CSTR volume needed to reach $X = 0.60$. Show the exit temperature, k, K_C and the equilibrium conversion at that temperature, and confirm that $X = 0.60$ is attainable.

(d) [6] (i) Show with a short calculation that no adiabatic reactor of any size can reach $X = 0.80$ with this feed. (ii) Sketch X versus T, showing the equilibrium curve $X_e(T)$ and the energy-balance line, and mark the adiabatic equilibrium point. *Do not calculate it.* Name two changes to the process that would raise the highest conversion that can be reached.

Problem 2: Reaction pathways and enzyme kinetics

[20 pts]

Part A: Fluorescence quenching. The fluorescence of a dye D in water falls when dissolved O_2 (Q) is present. The proposed mechanism is

(1) excitation:	$D \longrightarrow D^*$	rate $r_1 = I_a$ (constant for a fixed lamp)
(2) deactivation by solvent:	$D^* + M \xrightarrow{k_2} D + M$	
(3) quenching by O_2:	$D^* + Q \xrightarrow{k_3} D + Q$	
(4) fluorescence:	$D^* \xrightarrow{k_4} D + h\nu'$	

The measured light intensity is $I = k_4 C_{D^*}$. Let I_0 be the intensity without O_2.

(a) [7] Use the PSSH on D^* to show that $\dfrac{I_0}{I} = 1 + k' C_Q$, and express k' in terms of the rate constants. Explain why I_a does not appear in the result.

(b) [3] At $C_Q = 3.0$ mmol/L, $I_0/I = 2.5$. Find k', and the O_2 concentration at which the intensity falls to one quarter of I_0.

Part B: Enzyme kinetics. An enzyme (1.0 g/L) converts substrate S by Michaelis–Menten kinetics. Initial rates:

C_S (mmol/L)	1.0	2.0	4.0	8.0
$-r_S$ (mmol/L/min)	0.400	0.667	1.000	1.333

(c) [**4**] Use a Lineweaver–Burk analysis to find $V_{\max}$ and K_M (with units).

(d) [**6**] The same enzyme is used at **0.50 g/L** in a constant-volume batch reactor with $C_{S0} = 20$ mmol/L. Starting from the mole balance, derive the time needed to reach conversion X, and calculate the time for $X = 0.90$. State which Michaelis–Menten parameter changes with the enzyme loading, and why.

Problem 3: Cumene decomposition over Pt

[30 pts]

Cumene (C) decomposes to benzene (B) and propylene (P) over a Pt catalyst, $\mathrm{C} \longrightarrow \mathrm{B} + \mathrm{P}$. The proposed mechanism (single site; propylene goes directly to the gas and is not adsorbed) is

$$(1)\ \mathrm{C} + \mathrm{S} \rightleftharpoons \mathrm{C}\cdot\mathrm{S}\ (K_C) \qquad (2)\ \mathrm{C}\cdot\mathrm{S} \rightleftharpoons \mathrm{B}\cdot\mathrm{S} + \mathrm{P(g)}\ (K_S) \qquad (3)\ \mathrm{B}\cdot\mathrm{S} \rightleftharpoons \mathrm{B} + \mathrm{S}\ (K_B)$$

with overall equilibrium constant $K_P = K_C K_S / K_B$ and total site concentration C_t.

(a) [8] Derive the rate law $-r'_C$ assuming the surface reaction (2) is rate-limiting. Show the site balance.

(b) [6] If instead adsorption or desorption were rate-limiting, the rate laws would be

$$\text{adsorption-limited:} \quad -r'_C = \frac{C_t k_A \left(P_C - P_B P_P / K_P\right)}{1 + K_B P_B P_P / K_S + K_B P_B},$$

$$\text{desorption-limited:} \quad -r'_C = \frac{k_D C_t K_S K_C \left(P_C - P_B P_P / K_P\right)}{P_P + P_P K_C P_C + K_S K_C P_C}.$$

For a feed of pure cumene, write the *initial* rate $-r'_{C0}$ for all three rate-limiting steps, and sketch $-r'_{C0}$ against P_{C0} for each one on the same axes.

(c) [**8**] Initial rates measured with pure cumene are:

P_{C0} (atm)	0.5	1.0	2.0	4.0	8.0
$-r'_{C0}$ (mol/kg cat/s)	0.167	0.250	0.333	0.400	0.444

Which rate-limiting step is consistent with these data? Explain. Using a suitable linearisation, find the lumped rate constant k (with units) and K_C.

(d) [**8**] Pure cumene is fed at $F_{C0} = 10$ mol/s and $P_0 = 3$ atm to an isothermal fluidised-bed reactor that behaves as a CSTR, with no pressure drop. Under these conditions the reverse reaction is negligible. Benzene adsorbs with $K_B = 1.0$ atm^{-1}. Find the catalyst mass needed for $X = 0.50$. By what factor would you have underestimated the catalyst mass if you had ignored the adsorption of benzene? Why?

Problem 4: Chemical vapour deposition of germanium

[25 pts]

Germanium films are grown from $GeCl_4$ and H_2. The proposed mechanism is

gas-phase equilibrium: $GeCl_4(g) \rightleftharpoons GeCl_2(g) + Cl_2(g)$ (K_P)
adsorption (fast): $GeCl_2(g) + S \rightleftharpoons GeCl_2 \cdot S$ (K_A)
dissociative adsorption (fast): $H_2 + 2\,S \rightleftharpoons 2\,H \cdot S$ (K_H)
surface reaction (rate-limiting): $GeCl_2 \cdot S + 2\,H \cdot S \longrightarrow Ge(s) + 2\,HCl(g) + 2\,S$
with rate $r''_{Dep} = k_S f_{GeCl_2} f_H^2$

The surface is described by area fractions: vacant f_v, covered by $GeCl_2$ f_{GeCl_2}, and covered by H atoms f_H.

(a) [10] Derive the deposition rate r''_{Dep} in terms of P_{GeCl_2} and P_{H_2}.

(b) [5] Use the gas-phase equilibrium to rewrite your rate law in terms of P_{GeCl_4} and P_{Cl_2}. Explain why adding Cl_2 to the feed slows down the deposition.

(c) [6] At a fixed H_2 pressure with $\sqrt{K_H P_{H_2}} = 1.0$ and $K_A = 4.0$ atm^{-1}, find the $GeCl_2$ partial pressure that maximises the deposition rate. What is the apparent reaction order in $GeCl_2$ at very low and at very high P_{GeCl_2}?

(d) [4] Explain (i) why the denominator is *cubed*, (ii) why P_{H_2} appears under a square root, and (iii) why the total number of surface sites does not change as Ge is deposited.

Formula sheet (provided)

Steady-state energy balance	$\dot{Q} - \dot{W}_s - F_{A0}\sum_i \Theta_i C_{P,i}(T - T_{i0})$ $- \left[\Delta H^\circ_{Rx}(T_R) + \Delta C_P(T - T_R)\right] F_{A0} X = 0$
Heat of reaction	$\Delta H^\circ_{Rx} = \sum \nu_i H^\circ_i$ (per mole of A), $\quad \Delta C_P = \sum \nu_i C_{P,i}$
Arrhenius	$k(T) = k(T_1)\exp\left[\dfrac{E}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T}\right)\right]$
van 't Hoff ($\Delta C_P = 0$)	$K_C(T) = K_C(T_2)\exp\left[\dfrac{\Delta H^\circ_{Rx}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T}\right)\right]$
Design equations	batch (constant V): $\dfrac{dC_S}{dt} = r_S$ $\quad$ PBR: $F_{A0}\dfrac{dX}{dW} = -r'_A$ CSTR: $V = \dfrac{F_{A0}X}{-r_A}$, $\quad W = \dfrac{F_{A0}X}{-r'_A}$
Stoichiometry	liquid: $C_A = C_{A0}(1 - X)$, $\quad C_B = C_{A0}(\Theta_B + X)$ gas: $C_A = C_{A0}\dfrac{1 - X}{1 + \varepsilon X}\, y\dfrac{T_0}{T}$, $\quad P_A = P_{A0}\dfrac{1 - X}{1 + \varepsilon X}\, y$, $\quad \varepsilon = y_{A0}\delta$
PSSH	$r_{A^*} = \sum_i r_{iA^*} = 0$
Michaelis–Menten	$-r_S = \dfrac{V_{\max} C_S}{K_M + C_S}$, $\qquad V_{\max} = k_{\text{cat}} C_{E,t}$
Elementary surface steps	adsorption $r_{AD} = k_A\left(P_A C_v - \dfrac{C_{\text{AS}}}{K_A}\right)$; $\quad$ desorption $r_D = k_D\left(C_{\text{BS}} - K_B P_B C_v\right)$
Site / area balance	$C_t = C_v + \sum_i C_{i\cdot S} \qquad$ or $\qquad f_v + \sum_i f_i = 1$
Integral	$\displaystyle\int \frac{dx}{x} = \ln x$